Olympiad Test Paper — Social Science (Class 7) — Paper 4

Chapter 19: Infrastructure: Engine of India's Development

Paper 4 — (progressively harder)

Subject	Social Science (NCERT Class 7)
Chapter	19 — Infrastructure: Engine of India's Development
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: **+4** for each correct answer · **−1** for each wrong answer · **0** if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: invisible utilities (buried power grids, water pipelines, networks) and people-serving institutions (schools, hospitals, courts, parks) ARE infrastructure, the division being by function (physical vs social), not by cost or visibility; the British built the 1853 railways for extraction, trade and troop control, repurposed into a national lifeline only later; care for public infrastructure is a collective duty of citizens and government together, not the government's alone; J.C. Bose (c.1895) was a communication scientist who improved the coherer and chose not to patent much of his work, not a railway or highway builder; do not swap physical sub-types (transport vs utility vs communication vs energy) with one another or with social infrastructure; renewable energy parks, living root bridges and Kautilya's Arthashastra all sit INSIDE the chapter's scope. Do not write on this paper — mark answers on your answer sheet.

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1. A teacher writes on the board: 'A power grid you never see, a hospital you walk into, and a highway you drive on are all the same kind of thing.' The single idea this statement best captures is that:

(A) all three are physical infrastructure because all are built of concrete and steel
(B) only the highway is real infrastructure because it is the one you can both see and use
(C) all three are basic facilities a country needs to function, so all count as infrastructure despite their differences
(D) all three are social infrastructure because each is ultimately used by people

2. Two students debate why infrastructure is split into 'physical' and 'social' rather than 'big' and 'small'. The correct basis for the chapter's two-fold division is:

(A) the function it serves — systems that move and supply things versus services that build human capability

(B) the physical size of each project, with large works called physical and small ones social

(C) the year each was built, with older works called social and newer ones physical

(D) the material used, with concrete works called physical and brick ones social

3. Someone insists, 'If I cannot point my finger at it, it is not infrastructure.' The example from the chapter that most directly demolishes this is:

(A) a crowded sea port full of ships, which proves infrastructure is always visible

(B) buried water pipelines and underground power grids, which are unseen yet essential physical infrastructure

(C) a tall hospital building, which everyone in the city can clearly see

(D) a national highway stretching across the plains, plainly in open view

4. The Latin roots of 'infrastructure' are infra ('below') and structura ('a structure'). Using only these roots, the best literal reading is:

(A) a structure that is shared between two far-apart places

(B) the tallest structure that rises above a city skyline

(C) a structure put up in great haste and at low cost

(D) the underlying structure that lies beneath and supports everything built on top

5. A planner lists four works: a 4G mobile-tower network, a district court, a sewage pipeline, and a wind farm. How many of these four are PHYSICAL infrastructure?

(A) Two (B) All four

(C) Three (D) One

6. Which single statement combines two of the chapter's central ideas with no embedded error?

(A) infrastructure is the engine of development, and it spans both physical systems and social services

(B) infrastructure is the engine of development, but it covers only roads and buildings

(C) infrastructure is the final outcome of development, and it includes only social services

(D) infrastructure is the enemy of development, and it consists only of invisible systems

7. A student argues that whether something is infrastructure depends on how expensive it was to build. The clearest reason this is wrong is that:

(A) a cheap footpath and a costly expressway are both transport infrastructure, so cost does not decide the matter

(B) everything labelled infrastructure in the chapter costs exactly the same amount to build

(C) only the most expensive projects, such as ports and dams, are ever called infrastructure

(D) cheap works are counted as social infrastructure while costly works are counted as physical

8. Match each facility to its correct type: (i) a sewage-treatment plant and its pipes, (ii) a children's public park, (iii) a fibre-optic broadband network. The correct set is:

(A) (i) social, (ii) physical, (iii) physical communication

(B) (i) physical utility, (ii) physical, (iii) social

(C) (i) physical utility, (ii) social, (iii) physical communication

(D) (i) social, (ii) social, (iii) social

9. Of the following, the one that is NOT physical infrastructure is:

(A) a natural-gas pipeline supplying homes(B) a hydroelectric dam on a river

(C) a telecom satellite-uplink station (D) a district court complex

10. A school principal labels four items: (i) the school library, (ii) the road outside the gate, (iii) the electricity line powering the classrooms, (iv) the local court. How many are SOCIAL infrastructure?

(A) Three (B) Two

(C) One (D) Four

11. A new airport, a new public hospital, and a new water-supply line open in one week. Classified by the chapter's two arms, they are respectively:

(A) physical (transport), social, physical (utility)

(B) physical (transport), physical (utility), social

(C) social, physical (transport), physical (energy)

(D) physical (energy), social, physical (communication)

12. A facility supplies electricity to homes, factories, schools and hospitals alike. Its dual significance, per the chapter, is that it is:

(A) social infrastructure, because it serves people in their homes directly

(B) communication infrastructure, because it sends current as a signal to buildings

(C) physical energy/utility infrastructure that the other infrastructure depends on to operate

(D) not infrastructure at all, because private households also draw on it

13. A hospital and a power grid are very different, yet the chapter places both under 'infrastructure'. The correct reason is:

(A) both are physical infrastructure because both rely on electricity to work

(B) both are social infrastructure because both ultimately serve people

(C) both are basic facilities a country needs — the hospital as social, the grid as physical infrastructure

(D) neither is really infrastructure, because only roads count as infrastructure

14. Which item is a UTILITY rather than a social-infrastructure service?

(A) a government school
(B) a public hospital
(C) a district court
(D) piped water supply delivered to homes

15. Schools, hospitals, courts and parks are grouped together in the chapter. The common feature that places all four in one category is that all:

(A) are physical transport infrastructure that move goods between regions

(B) generate the nation's electricity for homes and factories

(C) carry wireless communication signals over long distances

(D) are social infrastructure aimed at people's wellbeing, learning, justice and recreation

16. A town has excellent roads, reliable power and a busy port, but almost no schools, hospitals or sanitation. Following the chapter's logic, this town will most likely:

(A) develop perfectly, since strong physical infrastructure alone guarantees progress

(B) struggle to develop fully, because development needs healthy, educated people as much as good physical systems

(C) be entirely unaffected, because social infrastructure plays no real role in development

(D) have to demolish its roads and port to make room for true development

17. Region X has reliable power, good highways and schools; Region Y has none of these. Other things equal, the chapter's reasoning predicts that:

(A) Y will develop faster, because it has far less infrastructure to maintain

(B) both will develop identically, since infrastructure does not affect development

(C) X is better placed to develop, because both its physical and social infrastructure support economic activity and human wellbeing

(D) X will fall behind, because having too much infrastructure slows a region down

18. The chapter calls infrastructure the 'backbone' of the economy. The most precise meaning of this metaphor is that:

(A) infrastructure is simply the single most expensive thing an economy owns

(B) infrastructure is rigid and can never be changed once it has been built

(C) like a backbone supporting a body, infrastructure carries and connects the economic activities that depend on transport, power, water and communication

(D) infrastructure is useful only to an economy's largest industries, not its small ones

19. A factory owner says good infrastructure cut his production costs. The chain of reasoning that best fits the chapter is:

(A) better roads and ports speed the movement of raw materials and finished goods, and reliable power keeps machines running, so production is cheaper and faster

(B) infrastructure raised his costs, because maintaining the roads near his factory is expensive for him

(C) infrastructure helped only by reducing the number of workers he had to employ

(D) infrastructure made no difference, because only the price of raw materials matters to a factory

20. Which statement correctly distinguishes how PHYSICAL and SOCIAL infrastructure each aid development?

(A) physical infrastructure produces educated people, while social infrastructure moves goods and power

(B) both kinds only move goods, and neither has any effect on people themselves

(C) physical infrastructure moves goods, power and information, while social infrastructure produces healthy, educated, capable people

(D) physical infrastructure helps cities only, while social infrastructure helps villages only

21. A hospital depends on a power grid to keep its equipment running. The relationship this illustrates is best described as:

(A) the hospital becoming energy infrastructure because it consumes a lot of electricity

(B) social infrastructure depending on physical energy infrastructure to function

(C) the power grid becoming social infrastructure because a hospital relies on it

(D) two unrelated facilities, since a hospital has no genuine need for power

22. An expressway and an ordinary village lane differ in several chapter-relevant ways. The expressway is best described as:

(A) a slow local road built and maintained by a single village panchayat

(B) a high-speed national road built by the central government to link cities across states

(C) an elevated urban rail line carrying commuters within one city

(D) a coastal route used by cargo ships travelling between sea ports

23. The Golden Quadrilateral links Delhi, Mumbai, Chennai and Kolkata. The name 'quadrilateral' is used because:

(A) it is built across exactly four small villages on the plains

(B) it connects four major cities, forming a four-sided road circuit across the country

(C) it has exactly four traffic lanes along its whole length

(D) it joins four foreign capital cities to one another

24. A goods company moves cargo from Delhi to Chennai. Using the chapter's framing, the Golden Quadrilateral helps it MOST by:

(A) providing a fast, direct highway link between major cities so goods reach distant markets quickly

(B) supplying electricity to the company's warehouses along the route

(C) treating the sewage produced at each of the company's offices

(D) granting the company a patent on the products it transports

25. NH44 is highlighted in the chapter. Which description best captures why it is significant among India's roads?

(A) it is a single city's metro line carrying daily commuters underground

(B) it is the sea port that handles the most containers in the country

(C) it is a large solar park feeding electricity into the national grid

(D) it is one of India's major national highways, notable as among the longest, knitting distant parts of the country together

26. Highways, railways, metros, airports and ports are all listed in the chapter. The single category that correctly contains ALL of them is:

(A) physical transport infrastructure (B) social infrastructure

(C) communication infrastructure (D) energy infrastructure

27. A district has fine roads and ports but suffers daily power cuts and broken water pipes. The chapter's reasoning suggests its development is held back mainly because:

(A) good roads and ports are themselves harmful and should be removed

(B) power and water have nothing to do with how transport or services run

(C) only the roads matter for development, so the failing utilities are irrelevant

(D) transport, factories and social services all depend on reliable power and water, which the failing utilities cannot provide

28. Two regions each have one missing piece. Region P lacks reliable electricity; Region Q lacks schools and hospitals. The chapter implies that, to develop:

(A) both should focus only on building more roads and ignore their actual gaps

(B) P should build more schools and Q should build more power stations

(C) P must strengthen its physical energy infrastructure while Q must strengthen its social infrastructure

(D) neither gap matters, because development depends on infrastructure not at all

29. The chapter says the British began India's railways in 1853. A student writes, 'so the railways have always been India's lifeline since 1853.' The most accurate correction is:

(A) the 1853 railways were first built for British colonial ends and only LATER became India's national lifeline

(B) correct — the railways carried ordinary Indian travellers as a lifeline from the very first day

(C) the railways were never a lifeline and still serve only British interests today

(D) the railways were built in 1853 by Indians for free public travel across the land

30. The British colonial motives for building the 1853 railways included all of the following EXCEPT:

(A) carrying India's raw materials to the ports for export

(B) giving Indian villagers free, fast travel purely for their own benefit

(C) distributing British-made goods across India to be sold

(D) moving troops quickly to keep control over the territory

31. The chapter uses the railways to teach a general principle about infrastructure. That principle is best stated as:

(A) infrastructure built by colonisers must always be torn down once they leave

(B) infrastructure can only ever serve the single purpose for which it was first built

(C) infrastructure built for one purpose can later be repurposed to serve a very different, often wider, need

(D) infrastructure built long ago becomes useless within just a few years

32. The railways carry about 20 million people each day. A student misreads this as '20 people a day'. The scale he has misjudged is by a factor of about:

(A) one hundred
(B) one thousand
(C) one million
(D) ten

33. Three dated facts appear in the chapter: the Indian Railways (1853), J.C. Bose's wireless work (about 1895), and Indian independence (1947). The chronologically correct order, earliest first, is:

(A) Indian Railways (1853), J.C. Bose's wireless work (1895), Indian independence (1947)

(B) J.C. Bose's wireless work (1895), Indian Railways (1853), independence (1947)

(C) independence (1947), Indian Railways (1853), Bose's wireless work (1895)

(D) Indian Railways (1853), independence (1947), Bose's wireless work (1895)

34. A history quiz claims, 'J.C. Bose designed and built India's first railway in 1853.' The single fact that makes this wrong is that:

(A) Bose built the railway, but in 1947 rather than in 1853

(B) the first railway was built by Indians for free travel, with Bose helping

(C) Bose was a communication scientist active around 1895, while the 1853 railways were a British colonial project, not his work

(D) Bose designed the Golden Quadrilateral rather than the railway

35. A region replaces its old goods-only colonial-era rail use with daily passenger services for millions. According to the chapter, this best illustrates that the railways:

(A) were always meant for Indian passengers from the moment they opened in 1853

(B) lost all usefulness once the British departed from the country

(C) remain a purely British-owned company carrying only freight today

(D) turned from a colonial extraction tool into a national lifeline serving ordinary Indians

36. A metro and the long-distance Indian Railways are both rail systems. The pair that correctly contrasts them is:

(A) a metro links distant states, while the Railways serve only one city

(B) a metro is a fast urban rail system easing city congestion, while the Railways form a nationwide network linking distant regions

(C) both are sea-transport systems built to carry cargo ships

(D) a metro is energy infrastructure, while the Railways are communication infrastructure

37. A city's roads are jammed every morning. The chapter's reasoning suggests a metro would help mainly because it:

(A) exports the city's raw materials to foreign sea ports

(B) generates renewable electricity to power the whole city

(C) carries cargo containers between the country's two coasts

(D) moves large numbers of commuters on rails, taking pressure off the congested roads

38. A student claims metros and expressways do the same job in the same way. The clearest distinction the chapter supports is that:

(A) a metro carries cargo ships, while an expressway carries aircraft

(B) a metro carries commuters on rails within a city, while an expressway is a high-speed road linking cities across states

(C) a metro is social infrastructure, while an expressway is energy infrastructure

(D) both are inter-state roads, differing only in the colour of their signs

39. The chapter notes 'aviation' comes from the Latin avis ('bird'). A consistent extension of this reasoning is that aviation infrastructure centres on:

(A) harbours and ships that carry cargo across the sea

(B) airports and aircraft that carry people and goods through the air

(C) power stations that generate electricity for the grid

(D) pipelines that carry clean water to people's homes

40. India is said to have about 159 airports. A student claims 'India therefore has exactly 159 sea ports too.' The correct response is:

(A) yes, airports and sea ports are always equal in number in any country

(B) no — 159 is the chapter's airport figure, and sea ports are a separate kind of coastal facility not given that same number

(C) no, because India actually has no airports at all

(D) no, because all 159 are in fact sea ports rather than airports

41. A logistics manager must ship containers overseas and also fly urgent parcels abroad. The two pieces of physical infrastructure she relies on are, respectively:

(A) a sea port for the containers and an airport for the urgent parcels

(B) a power grid for the containers and a metro for the parcels

(C) a school for the containers and a hospital for the parcels

(D) a water pipeline for the containers and a court for the parcels

42. Port A handles 5,000 TEUs in a day and Port B handles 8,000 TEUs the same day. Using the meaning of TEU, the correct interpretation is:

(A) Port B handled 3,000 more standard twenty-foot-equivalent containers that day, indicating greater container-handling capacity

(B) Port B ran its trains 3,000 kilometres farther that day than Port A

(C) Port A is 3,000 metres taller in structure than Port B

(D) Port B flew 3,000 more aircraft that day than Port A

43. Why is a standard unit like the TEU useful for comparing the world's ports?

(A) it measures how fast metros run inside a port city

(B) it gives a common container measure, so ports of different sizes and cargo can be compared fairly

(C) it records the exact height of a port's tallest crane

(D) it counts how many airports each port happens to own

44. India's major sea ports lie on both its coasts. The geographic reason this matters for trade is that:

(A) coastal ports are useless, because ships can never reach the country's interior

(B) ports must instead be in the Himalayas to handle the heaviest cargo

(C) ports on both the eastern and western coasts give India sea access in two directions for importing and exporting goods

(D) ports on both coasts automatically double the country's number of airports

45. Someone claims, 'All modern communication technology, including radio, was invented entirely in the West.' Drawing on the chapter, the strongest counter is:

(A) India only began studying radio after independence in 1947

(B) radio waves were first used in ancient India for long-distance trade

(C) the British invented radio when they built the railways in 1853

(D) the Indian scientist J.C. Bose demonstrated radio-wave communication and improved the 'coherer' detector, showing India contributed to the science of communication

46. A 'coherer', improved by J.C. Bose, played which role in early wireless communication?

(A) it generated the electricity that powered the early railways

(B) it detected and decoded the transmitted wireless (radio) signals at the receiving end

(C) it pumped clean water into the homes of a growing city

(D) it carried passengers along an elevated urban rail track

47. The chapter notes J.C. Bose 'chose not to patent much of his work.' Given that a patent grants an exclusive right, the most reasonable inference is that Bose:

(A) believed his own inventions did not work and were worthless

(B) was legally forbidden by the British from ever holding any patent

(C) was willing to let his scientific ideas be freely used rather than seek personal exclusive profit from them

(D) wanted to keep all of his findings secret from everyone forever

48. If Bose HAD patented every invention, the chapter's definition of a patent tells us the most direct consequence would have been:

(A) his inventions would automatically have stopped working

(B) for a number of years, only he or those he permitted could legally make, sell or use those inventions

(C) everyone else could have copied his inventions freely with no limit

(D) the British would have been forced to leave India much sooner

49. Telecommunication comes from tele ('far') and communicare ('to share'). By this root meaning, which everyday activity is an example of telecommunication?

(A) driving a car along a national highway between two cities

(B) loading shipping containers onto a vessel at a sea port

(C) generating electricity from the wind at a coastal wind farm

(D) speaking to a relative in another city over the telephone

50. Communication infrastructure and transport infrastructure both 'connect' places, yet differ. The clearest distinction the chapter draws is that:

(A) communication infrastructure moves information across distance, while transport infrastructure moves people and goods

(B) communication moves goods, while transport moves only information

(C) both move only electricity and nothing else between places

(D) communication works only abroad, while transport works only inside India

51. Energy infrastructure includes dams, solar parks and wind farms. Sorting these by their energy source, the correct statement is:

(A) all three burn coal to produce their energy

(B) only the dam is energy infrastructure, while the others are social

(C) none of them is infrastructure, because nature supplies the energy free

(D) all three can supply renewable energy — water for the dam, sunlight for the solar park, and wind for the wind farm

52. A city replaces a coal power plant with a large solar park. In the chapter's terms, this change makes the city's energy infrastructure more:

(A) sustainable, because it draws on a cleaner, renewable source with less environmental harm

(B) polluting, because solar parks burn more fuel than coal plants do

(C) social rather than physical infrastructure

(D) useless, because renewable energy cannot power a whole city

53. The chapter argues that energy and utilities matter to ALL other infrastructure. The reasoning behind this is that:

(A) energy and utilities are used only inside private homes and nowhere else

(B) energy and utilities entirely replace the need for roads and schools

(C) energy and utilities are completely unrelated to any other infrastructure

(D) transport, communication and social services all need power and water to function, so energy and utilities underpin them

54. A 'utility' takes its name from the Latin utilis ('useful'). The reason this name suits utilities like water, electricity and gas is that they:

(A) are the most beautiful structures built in any city

(B) are the fastest roads available for long-distance travel

(C) are devices that detect and decode wireless signals

(D) are basic useful services supplied to homes and workplaces for everyday needs

55. Sanitation is listed under social infrastructure. The strongest reason it belongs there rather than being ignored is that:

(A) clean sanitation protects people's health and wellbeing, a core aim of social infrastructure

(B) sanitation makes aircraft take off and fly faster

(C) sanitation increases the container count handled at a port

(D) sanitation adds extra kilometres to the national highways

56. The chapter links social infrastructure to development through people. The most complete version of this link is:

(A) social infrastructure speeds up cargo trains, raising the country's exports

(B) social infrastructure replaces the need for any physical infrastructure at all

(C) social infrastructure benefits only the people who actually build it

(D) schools and hospitals produce educated and healthy citizens, who can then work, invent and contribute to the country's growth

57. A village treats its hand-pump, the school building and the approach road as 'everyone's to look after'. The civic principle this reflects is:

(A) private ownership, where each facility belongs to a single family

(B) government monopoly, where citizens are barred from touching any of it

(C) collective responsibility — shared public infrastructure is cared for jointly by the community

(D) foreign maintenance, where outsiders are expected to care for all of it

58. Two students argue about a vandalised public bus shelter. Student 1 says, 'Only the government must fix and protect it.' Student 2 says, 'Citizens share that duty too.' The chapter supports:

(A) Student 1, because citizens have no role at all in public property

(B) neither, because public infrastructure needs no care whatsoever

(C) Student 1, because only foreigners are meant to maintain Indian infrastructure

(D) Student 2, because caring for public infrastructure is a collective responsibility shared by citizens and government

59. Which set of citizen actions best demonstrates collective responsibility for shared infrastructure?

(A) breaking the taps in a public washroom when no one is watching

(B) using a public park properly, not damaging benches, and reporting a broken streetlight

(C) ignoring a leaking public pipeline because it is 'not my job'

(D) waiting for a foreign agency to come and repair the local road

60. A student summary reads: 'Infrastructure is only visible roads and buildings, the British built the 1853 railways to help Indians, and caring for it is the government's job alone.' Reviewing the whole chapter, the single FULLY correct statement is instead:

(A) infrastructure, the engine of development, has physical and social arms — including invisible utilities and natural works like Meghalaya's living root bridges — and is best kept working through care shared between government and citizens

(B) infrastructure is only physical, and the 1853 British railways were built to benefit Indian travellers

(C) infrastructure means only invisible systems, and its upkeep is solely the government's responsibility

(D) infrastructure includes social services, but J.C. Bose built India's first railway in 1853

Solutions — Social Science (Class 7), Chapter 19: Infrastructure: Engine of India's Development — Paper 4

Paper 4 — (progressively harder)

Marking: **+4** correct, **−1** wrong, **0** unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	A	21	B	31	C	41	A	51	D
2	A	12	C	22	B	32	C	42	A	52	A
3	B	13	C	23	B	33	A	43	B	53	D
4	D	14	D	24	A	34	C	44	C	54	D
5	C	15	D	25	D	35	D	45	D	55	A
6	A	16	B	26	A	36	B	46	B	56	D
7	A	17	C	27	D	37	D	47	C	57	C
8	C	18	C	28	C	38	B	48	B	58	D
9	D	19	A	29	A	39	B	49	D	59	B
10	B	20	C	30	B	40	B	50	A	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) The unifying idea is that infrastructure means the basic facilities a country needs to function, which embraces a hidden utility, a people-serving institution and a transport route alike. Calling all three physical misclassifies the hospital, which is social. Restricting it to the visible highway is the visibility error, since invisible systems still qualify. Calling all three social misclassifies the grid and the highway, which are physical.

2. (A) The chapter divides infrastructure by function: physical systems carry goods, power, water and information, while social services like schools and hospitals build people's capability. Size, age and building material are not the dividing axis, and a small clinic is still social while a small footpath is still physical, so none of those bases fits.

3. (B) The claim targets invisibility, so only an invisible-yet-essential example refutes it: buried pipelines and underground grids are vital physical infrastructure that cannot be seen. The port, the hospital building and the highway are all highly visible, so pointing to them concedes rather than rebuts the claim.

4. (D) Infra means 'below' and structura 'a structure', giving the underlying base that supports what rests on it. 'Shared between far places' belongs to the tele/communicare roots of telecommunication, not infrastructure. 'Tallest above a city' inverts infra into 'above'. 'Built in haste at low cost' invents a speed-and-price meaning the roots do not carry.

5. (C) Physical infrastructure spans transport, utilities, communication and energy. The mobile-tower network (communication), the sewage pipeline (utility) and the wind farm (energy) are all physical, giving three. The district court is a social-infrastructure institution, so it is excluded. 'Two' or 'one' undercount the physical works, and 'all four' wrongly treats the court as physical.

6. (A) The chapter calls infrastructure the engine of development and gives it two arms, physical and social; only the correct option states both accurately. Limiting it to 'roads and buildings' or 'only social services' repeats a misconception, and 'enemy/final outcome' misstates its driving role.

7. (A) Infrastructure is defined by function, not price, so both a cheap footpath and a costly expressway are transport infrastructure. The claim that all infrastructure costs the same is false, that only the priciest works count denies cheap roads, and that cost decides physical-versus-social is the very confusion the chapter warns against.

8. (C) A treatment plant and its pipes are physical-utility hardware; a public park is a social-infrastructure facility; a fibre-optic network is physical communication. The other sets swap the park and the network or lump everything as social, misclassifying the utility plant or the communication network.

9. (D) A court is a social-infrastructure institution, so it is the odd one out. A gas pipeline is a physical utility, a hydroelectric dam is physical energy, and a satellite-uplink station is physical communication — all three are physical.

10. (B) Social infrastructure serves people's wellbeing, learning and justice: the library (part of the school) and the court qualify, giving two. The road is physical transport and the electricity line is physical energy, so neither counts. 'One' misses a social item, while 'three' or 'four' wrongly count the road or power line as social.

11. (A) An airport is physical transport, a hospital is social infrastructure, and a water-supply line is a physical utility. The distractors mislabel the hospital as a utility, the airport as social, or the water line as energy or communication, each breaking the correct sub-type.

12. (C) Electricity supply is physical energy/utility infrastructure, and because transport, communication and social services all need power, it underpins them. Calling it social confuses 'serves people' with the functional divide, calling it communication confuses carrying power with carrying information, and private household use does not strip a public supply system of being infrastructure.

13. (C) Both qualify as basic facilities a country needs, the hospital under the social arm and the grid under the physical arm. Saying both are physical misclassifies the hospital, saying both are social misclassifies the grid, and 'only roads count' is the misconception the chapter rejects.

14. (D) A utility is a basic supplied service such as water, electricity or gas, so piped water supply fits exactly. A school, a hospital and a court are social-infrastructure institutions, not utilities, so they are wrong for 'utility'.

15. (D) Schools, hospitals, courts and parks all serve human needs — learning, health, justice and recreation — which is the aim of social infrastructure. None moves goods (transport), generates electricity (energy) or carries signals (communication), so the other options misclassify the whole group.

16. (B) The chapter stresses that physical and social infrastructure are two arms that work together; strong physical systems without healthy, educated people leave development incomplete. 'Physical alone guarantees progress' and 'social plays no role' both deny the social arm, and tearing down working physical infrastructure would harm, not help, development.

17. (C) X's combination of physical systems and social services gives it the foundation for both economic activity and human wellbeing — the backbone idea. 'Y faster with less to maintain' and 'too much slows X' invert the chapter's claim, and 'infrastructure does not affect development' contradicts its central thesis.

18. (C) The backbone image is about support and connection: economic activity rests on and is linked by transport, power, water and communication. It is not about being the costliest asset, about rigidity, or about serving only large industries, so those readings mistake a structural-support image for cost, inflexibility or selectivity.

19. (A) Transport infrastructure moves inputs and outputs quickly while energy infrastructure powers the machines, lowering both cost and time — the multi-link cause the chapter supports. The distractors claim infrastructure raises a factory's costs, reduce its benefit to cutting headcount, or deny any effect, all contrary to the backbone-of-the-economy idea.

20. (C) Physical systems handle the flow of goods, energy and information; social services build human capability — the two complementary roles. The first distractor swaps them, 'both only move goods' erases the human-development role, and 'city-only versus village-only' invents a geographic split the chapter does not make.

21. (B) A hospital is social infrastructure that relies on physical energy infrastructure to operate, showing the layers depend on one another. Consuming electricity does not turn the hospital into energy infrastructure, a hospital's use does not make the grid social, and hospitals plainly do need power.

22. (B) An expressway is a high-speed inter-state national road built by the central government, the defining contrast with a village lane. A village-maintained local road is its opposite, an elevated urban rail line is a metro, and a coastal cargo route is sea transport — none is a road expressway.

23. (B) 'Quadrilateral' means four-sided; the network joins four major Indian cities into a four-sided circuit. It is not a four-village project, the name does not refer to lane count, and the four cities are Indian, not foreign capitals.

24. (A) The Golden Quadrilateral is transport infrastructure; its benefit to a goods company is rapid movement between major cities. Supplying electricity is energy infrastructure, treating sewage is a utility function, and granting a patent is a legal right over an invention — none is what a highway network provides.

25. (D) NH44 is a major and notably long national highway connecting far-apart regions. The other options describe entirely different infrastructure types — a metro (urban rail), a port (sea transport) and a solar park (energy) — none of which is a national highway.

26. (A) Roads, rail, metros, airports and ports all carry people or goods, so they are all physical transport infrastructure. They are not social services, not communication systems (which carry information), and not energy systems (which carry power).

27. (D) Energy and utilities are a base layer beneath the rest: when power and water fail, the transport, factories and services that rely on them also falter, so the district is held back. Good roads and ports are not harmful, power and water are closely tied to how other infrastructure runs, and the failing utilities are far from irrelevant.

28. (C) P's gap is in physical energy infrastructure and Q's is in social infrastructure, so each must close its own gap. Ignoring the real gaps to build roads does not fix either, swapping the fixes addresses neither region's actual shortfall, and 'infrastructure does not matter' contradicts the chapter's thesis.

29. (A) The chapter's key point is a change of purpose: built in 1853 for colonial extraction and control, the railways became India's lifeline only afterward. Saying they served Indians 'from the first day' keeps the colonial-benefit myth, 'never a lifeline / still British' denies their present role, and 'built by Indians for free travel in 1853' is historically false.

30. (B) Export of raw materials, sale of British goods and rapid troop movement were the real colonial motives, so they are not exceptions. The one that does NOT belong is 'free fast travel for villagers' own benefit', which is precisely the myth the chapter corrects.

31. (C) Railways built for colonial gain became a national lifeline, showing infrastructure can be repurposed for new and broader ends. 'Must be torn down', 'only one purpose', and 'useless within years' each contradict this lesson of adaptive, lasting value.

32. (C) Twenty million divided by twenty equals one million, so reading '20 million' as '20' understates the figure by roughly a factor of a million. Factors of a hundred, a thousand or ten are all far too small to span the gap between 20 and 20,000,000.

33. (A) The railways came in 1853, Bose's wireless work about 1895, and independence in 1947, so 1853 < 1895 < 1947 is the right order. The other sequences place 1895 before 1853, or 1947 before 1853, or Bose's 1895 work after 1947, all violating the actual dates.

34. (C) Bose's contribution was to radio-wave communication around 1895; the 1853 railways were a British colonial undertaking, so crediting him with building them is doubly wrong on both person and project. Shifting the date to 1947, calling it a free Indian project, or swapping in the Golden Quadrilateral all keep the false attribution to Bose.

35. (D) The shift from colonial freight to mass passenger service is exactly the repurposing the chapter describes — a colonial tool becoming a national lifeline. 'Always for Indians from 1853' denies the colonial origin, 'lost all usefulness' is the opposite of becoming a lifeline, and 'purely British-owned freight only' contradicts the railways' present role.

36. (B) Metros are intra-city rapid transit that eases congestion, while the Indian Railways span the whole country between regions. The first distractor swaps their scopes, both are rail not sea transport, and both are physical transport, not energy or communication infrastructure.

37. (D) A metro's distinctive value is mass urban commuting that reduces road congestion, exactly the city's problem. Exporting raw materials is colonial-era freight rail or ports, generating electricity is energy infrastructure, and moving containers between coasts is sea/port transport — none is the metro's role.

38. (B) A metro is urban rail for commuters; an expressway is an inter-state high-speed road — same transport family, different role and mode. Ships and aircraft belong to sea and air transport, both are physical transport (not social or energy), and a metro is not a road, so 'both are inter-state roads' is false.

39. (B) Avis ('bird') points to flight, so aviation infrastructure is airports and aircraft moving through the air. Harbours and ships are sea transport, power stations are energy, and water pipelines are utilities — none matches the 'bird'/air meaning.

40. (B) The figure 159 refers to airports; sea ports are different coastal facilities, and the chapter does not equate the two counts. 'Always equal' is an unwarranted leap, 'no airports' contradicts the figure, and 'all 159 are sea ports' misattributes the number.

41. (A) Container shipping uses a sea port and fast air freight uses an airport, both physical transport suited to each task. Power grids and metros do not move overseas cargo or parcels, schools and hospitals are social services, and pipelines and courts are utilities and social facilities — none ships goods abroad.

42. (A) TEU (Twenty-foot Equivalent Unit) counts standard shipping containers, so 8,000 – 5,000 = 3,000 more containers at Port B, showing larger throughput. TEU is not a measure of train distance, of height, or of aircraft, so the other readings misuse the unit.

43. (B) A standard equivalent unit lets very different ports be compared on the same scale of container throughput. TEU has nothing to do with metro speed, crane height or airport ownership, so those options misidentify what the unit standardises.

44. (C) Ports on both coasts open trade routes in two directions, widening import-export access. Coastal ports are not useless for trade, the landlocked Himalayas have no sea access for ports, and coastal ports do not multiply airports — those options confuse port geography with other facts.

45. (D) Bose's work on radio-wave communication and the coherer is the chapter's evidence that India contributed to communication science, rebutting the 'entirely Western' claim. 'Only after 1947' and 'ancient trade use' are unsupported, and tying radio to the 1853 railways confuses two unrelated facts.

46. (B) The coherer was an early detector that picked up and decoded wireless signals, which Bose improved. It did not generate railway power, pump water (a utility function) or carry passengers (a transport function), so those options confuse communication with energy, utilities and transport.

47. (C) Declining patents meant Bose did not lock up his ideas for personal exclusive gain, leaving them open for others' use. It does not imply the work was worthless, that he was legally barred, or that he kept everything secret — declining a patent is the opposite of hiding work away.

48. (B) A patent grants the holder the sole right to make, sell or use the invention for a set period, so patenting would have legally restricted others for those years. Patenting does not affect whether devices function, 'everyone could copy freely' is the opposite of what a patent does, and it has nothing to do with ending colonial rule.

49. (D) Tele ('far') plus communicare ('to share') means sharing information over distance, which a long-distance phone call exactly does. Driving on a highway is transport, loading containers is sea transport, and generating power is energy — none of these shares information at a distance.

50. (A) The essential split is what each moves: communication carries information, transport carries people and goods. The first distractor swaps these roles, 'both move only electricity' describes energy rather than these two, and 'communication abroad only / transport India only' invents a false geographic divide.

51. (D) Hydro (dams), solar parks and wind farms all draw on renewable natural sources — water, sun and wind. None of them burns coal, all three are energy infrastructure (not social), and using a free natural source does not stop the built systems from being infrastructure.

52. (A) Switching to solar uses a clean, renewable source, reducing pollution — the definition of more sustainable infrastructure. Solar does not burn fuel (so it is not more polluting), it remains physical energy infrastructure (not social), and renewables are not useless for cities.

53. (D) Trains, phones, hospitals and schools all depend on power and water, so energy and utilities form a base layer beneath the rest. They are not home-only, they support rather than replace roads and schools, and they are far from unrelated to other infrastructure.

54. (D) Utilis means 'useful', and utilities are the basic useful services — water, electricity, gas — supplied for daily needs. They are not defined by beauty, they are not roads (that is transport), and a signal-decoding device is the coherer, a communication device, not a utility.

55. (A) Sanitation safeguards health and wellbeing, the central purpose of social infrastructure, so it is grouped there. It has no bearing on aircraft speed, port throughput or highway length, so those options confuse a health measure with unrelated transport facts.

56. (D) Social infrastructure develops human capability — health and education — which feeds back into national growth. It does not speed cargo (that is transport), cannot replace physical infrastructure, and benefits society broadly, not just its builders.

57. (C) Treating shared facilities as everyone's to maintain is collective responsibility, the chapter's civic ideal. It is not private family ownership, not a government monopoly that bars citizens, and not reliance on outsiders — those misread the shared-duty idea.

58. (D) The chapter teaches that maintaining public infrastructure is a collective duty of citizens as well as the government, so Student 2 is right. Student 1's 'government-only' view is the misconception corrected, 'needs no care' is false, and 'only foreigners maintain it' is unsupported.

59. (B) Proper use, avoiding damage and reporting faults are exactly the citizen behaviours the chapter calls collective responsibility. Breaking taps and ignoring a leak harm shared infrastructure, and waiting on a foreign agency abdicates the shared duty — all three are the opposite of responsible care.

60. (A) The correct statement combines several chapter ideas accurately: infrastructure drives development, has physical and social arms (including invisible utilities and sustainable natural works such as the Khasi and Jaintia living root bridges, an idea with ancient roots in Kautilya's Arthashastra), and is sustained by shared care. Each distractor hides an error — denying the social arm and the true colonial railway motive, restricting infrastructure to invisible systems with government-only care, or falsely crediting J.C. Bose, a communication scientist of about 1895, with building the 1853 railway.